s10_vol_rank_tilt_levered

equity strategy | family: equity-volatility | status: research | derived from `s10_vol_rank_tilt`

What this strategy does

s10 with the lower volatility spent on 1.67x base unit and a 4x gross-leverage cap

Why it should work. PEAD alpha per dollar is highest in calm stocks; rank-weight toward them without depending on vol levels.

Design notes (from the strategy's source file).

Strategy 10 -- Strategy 6 tilted toward calm stocks by volatility RANK (research).

Strategy 9b sized each event by 1/idiosyncratic-vol (Sharpe 1.30 -> 1.47). The attribution work showed that PEAD alpha per dollar is concentrated in low-volatility names and almost absent in the wildest tercile, and that the 1/vol weight, being linear in the vol LEVEL, only mildly demotes the high-vol tail. Weighting by (1.02 - vol percentile)^2 -- where the percentile is measured against the previous quarter's cross-section within the stock's size quintile -- demotes it hard while staying smooth:

Sharpe 1.30 (S6) -> 1.47 (9b) -> 1.81; max drawdown -16.6% -> -11.6%; every era improves split-sample check: exponent picked on 1996-2004 data alone (best k ~ 3) lifts 2005-2013 Sharpe 0.93 -> 1.10, the same as the full-sample best, so the gain is not an artifact of one period.

Lower volatility buys room: `s10_vol_rank_tilt_levered` spends it on 1.7x the base unit and a 4x gross cap for ~9% a year at ~5% vol (a market-neutral book clearing the S&P 500's return in this sample). More sizing than that stops paying: the 5%-of-ADV liquidity cap starts to bind and 2008-13 Sharpe decays.

STATUS research: the exponent and sizing were chosen after seeing 9b's full-sample results (the split-sample check above is the only out-of-sample evidence), and the levered variant leans on small-cap capacity.

parameter	value
base_unit_fraction	0.005
max_gross_leverage	4.0
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

metric	s10 vol rank tilt levered	market (CRSP VW)
Annualized return	9.23%	9.94%
Annualized volatility	5.17%	19.35%
Sharpe (NAV-pct)	1.73	0.59
Sortino	2.64	
Max drawdown	-15.9%	-55.5%

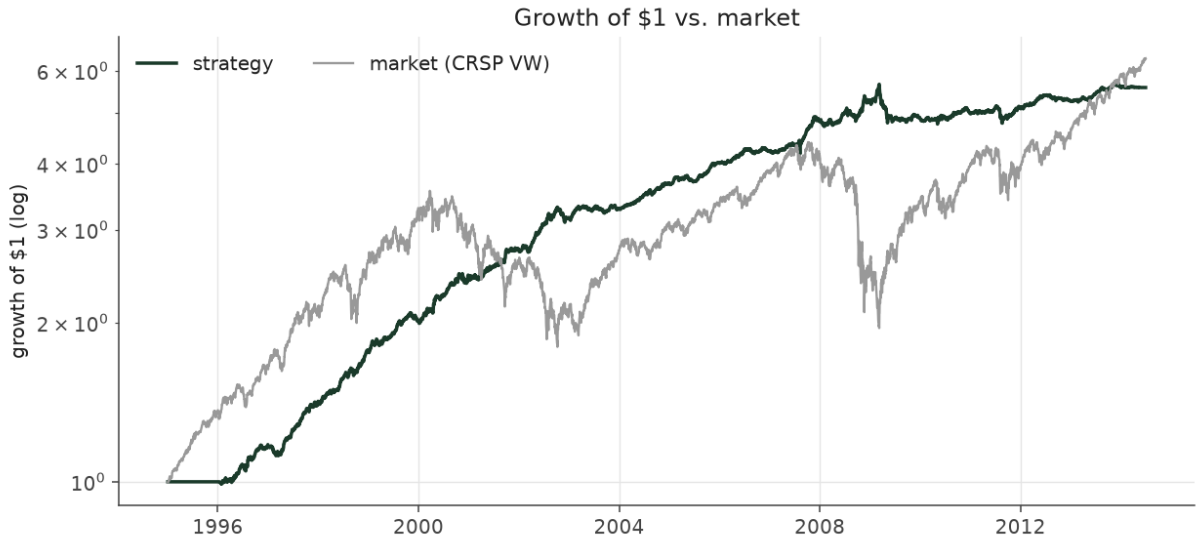
Calmar	0.58	
Final NAV ($10M start)	$55,789,745	
Worst day	-3.60%	
% positive days	52.5%	
Sharpe 1996-2001	2.86	
Sharpe 2002-2007	1.96	
Sharpe 2008-2013	0.50	

Correlation of daily returns with the market: **-0.03**.

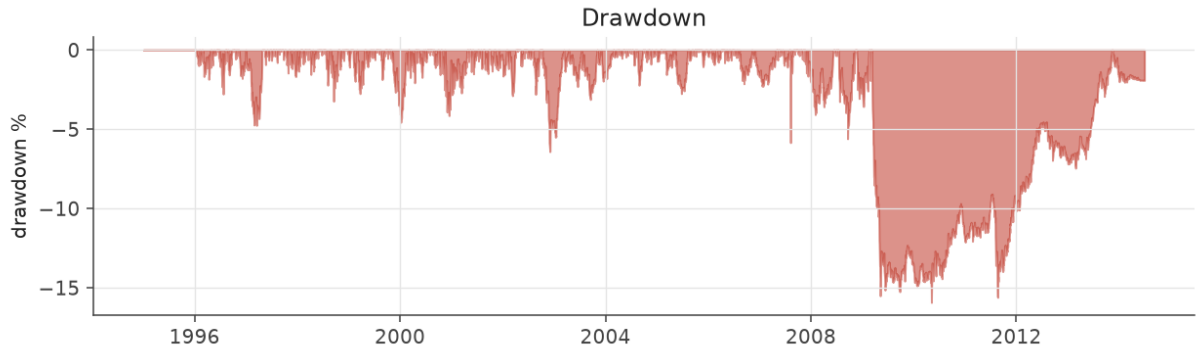
• best Sharpe among equity: `s10_vol_rank_tilt` (1.81); this one: 1.73

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0923

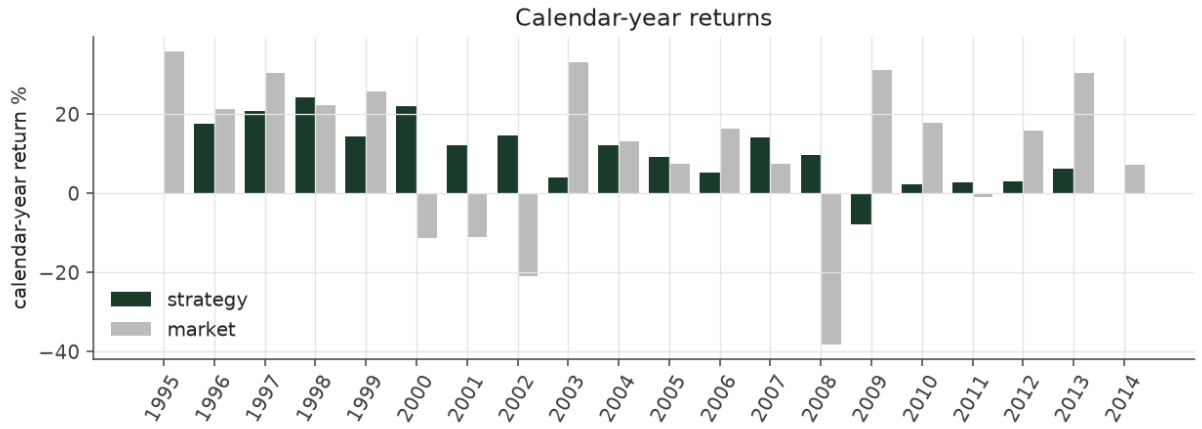
• best Calmar among equity: `s10_vol_rank_tilt` (0.614); this one: 0.579



NAV (log scale)



Drawdown from running peak

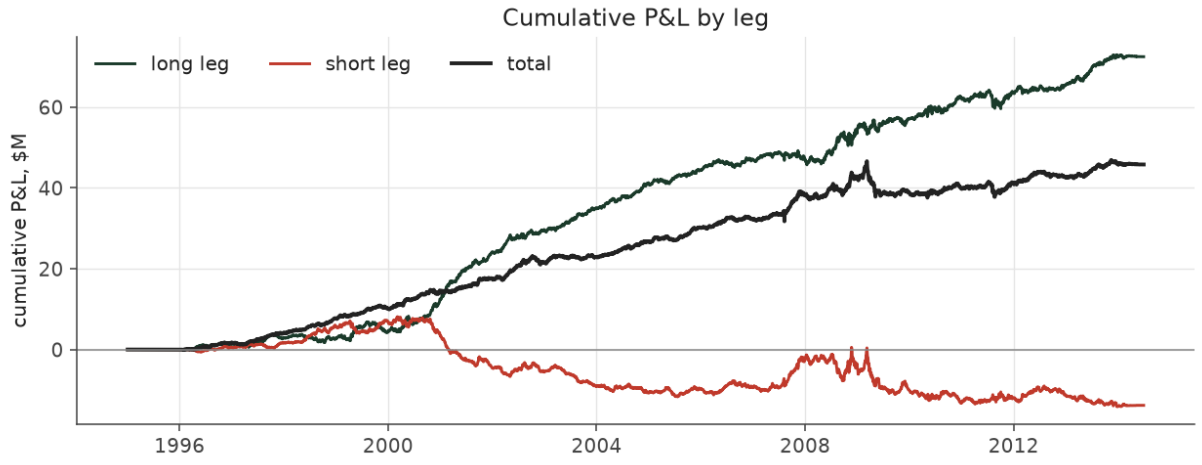


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	72.46	-13.81
P&L after costs ($M)	66.24	-20.45
of which common (average-event) component ($M)	55.30	-54.77

of which excess / PEAD component ($M)	17.16	40.96
excess minus costs ($M)	10.94	34.32
gross return per $ traded (bp)	208.91	-39.14
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.45	-0.16

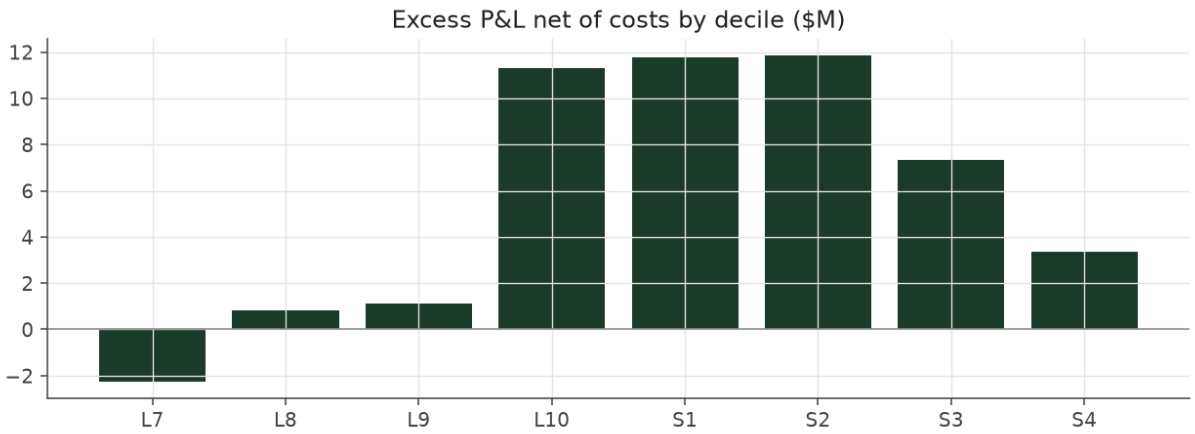
Daily P&L correlation between the two legs: **-0.77**.

• **The short leg loses money in absolute terms** (-13.8M before costs) while the long leg earns +72.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 55.3M / 54.8M. Stripping that common component, the surprise signal itself earned +17.2M on the long side and +41.0M on the short side.

• The two legs' daily P&L are -0.77 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.45 long / -0.16 short vs 1.73 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	326.55	3.79	-1.68	0.59	-2.27	-5.0%	-69.55
long	8	16,588	843.87	14.71	2.34	1.52	0.81	1.8%	9.64
long	9	17,747	1,080.59	17.10	3.02	1.91	1.11	2.5%	10.29
long	10	17,174	1,217.60	30.64	13.48	2.20	11.29	24.9%	92.70
short	1	17,169	1,171.25	-4.34	14.08	2.29	11.79	26.0%	100.64
short	2	17,281	1,084.11	-6.66	13.88	2.02	11.85	26.2%	109.34
short	3	15,970	888.83	-6.83	8.96	1.64	7.32	16.2%	82.30
short	4	9,171	384.55	-2.61	4.04	0.68	3.36	7.4%	87.37



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	343.64	5.44	7.65	1.29	6.36	14.0%	184.96
2	22,885	1,016.63	8.45	11.48	2.83	8.65	19.1%	85.09

3	25,501	1,544.72	12.48	14.91	3.26	11.65	25.7%	75.44
4	27,583	1,953.20	11.43	14.10	3.12	10.98	24.3%	56.24
5	28,947	2,139.15	7.99	9.98	2.37	7.62	16.8%	35.60

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	251.16	5.02	6.70	1.26	5.45	12.0%	216.90
2	23,084	1,138.52	8.26	13.18	3.42	9.77	21.6%	85.78
3	25,851	1,809.17	14.81	18.74	3.62	15.12	33.4%	83.59
4	27,665	1,938.19	9.63	10.53	2.71	7.82	17.3%	40.34
5	28,874	1,860.06	8.04	8.93	1.86	7.07	15.6%	37.98

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	2,914.53	2.37	17.60	4.88	12.71	28.1%	43.62
41-60d	25,652	1,600.35	11.07	9.58	2.99	6.59	14.6%	41.18
61-90d	35,324	1,883.42	13.71	15.21	3.65	11.56	25.5%	61.38
>90d	12,298	599.04	18.64	15.73	1.34	14.39	31.8%	240.28

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	1,122.39	17.52	19.70	2.06	17.64	39.0%	157.16
2002-07	39,956	2,449.43	20.47	24.04	4.43	19.61	43.3%	80.07
2008-13	45,278	3,425.53	7.80	14.37	6.37	8.01	17.7%	23.37

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	729.85	5.33	6.37	1.46	4.91	10.9%	67.28
Chems	2,590	441.60	3.43	4.07	0.69	3.38	7.5%	76.53
Durbl	2,843	460.22	2.24	2.98	0.74	2.23	4.9%	48.49
Enrgy	6,194	534.68	5.92	7.08	0.90	6.18	13.7%	115.58
Hlth	11,696	673.29	3.17	4.07	1.28	2.79	6.2%	41.47
Manuf	12,034	655.21	3.13	4.54	1.24	3.30	7.3%	50.41
Money	17,583	657.72	5.56	7.08	1.36	5.72	12.6%	86.93
NoDur	5,081	542.02	2.69	3.39	0.92	2.47	5.4%	45.49
Other	18,377	706.61	5.30	6.85	1.47	5.38	11.9%	76.12
Shops	11,984	672.08	4.85	5.88	1.28	4.61	10.2%	68.53

Telcm	3,981	466.01	2.81	3.57	0.75	2.83	6.2%	60.67
Utils	4,672	458.03	1.34	2.24	0.78	1.47	3.2%	32.00

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	625.15	5.22	6.78	1.11	5.66	12.5%	90.58
low vol	40,091	3,855.29	24.00	31.48	7.20	24.28	53.6%	62.98
mid vol	37,903	2,416.18	15.81	19.07	4.36	14.71	32.5%	60.90

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	9.23%	0.00%
0.5%	3.4	8.87%	0.36%
1.0%	6.9	8.49%	0.74%
2.0%	13.8	7.65%	1.58%
5.0%	34.4	3.97%	5.25%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.